Test2 Review, MAT 1575 Professor Chiu

1. Evaluate the following indefinite integrals:

$$(a)\int \sin^3(x)\,dx \qquad (b)\int \sin^5(x)\cos^2(x)\,dx \qquad (c)\int \sec^4(x)\,dx \qquad (d)\int \sin^2(x)\cos^2(x)\,dx$$

2. Evaluate the following indefinite integrals:

$$(a)\int \frac{dx}{\sqrt{1+9x^2}} \qquad (b)\int \frac{dx}{x^2\sqrt{1-x^2}} \qquad (c)\int \frac{dx}{x^2\sqrt{1+x^2}}$$

3. Evaluate the following indefinite integrals:

$$(a)\int \frac{2x^2+4x+12}{x^2+2x+5}dx \qquad (b)\int \frac{3x+2}{x^2+2x-8}dx \qquad (c)\int \frac{3x+4}{(x^2+4)(3-x)}dx$$

4. Evaluate the following improper integrals:

$$(a)\int_0^{\infty} \frac{5}{\sqrt[3]{x+5}}dx \qquad (b)\int_0^{1} \frac{\ln(x)}{\sqrt{x}}dx \qquad (c)\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$$

5.Find the Taylor polynomial of degree 2 for the given function, centered at the given number a:

$$(a) f(x) = e^x \ at\ a = 1 \qquad (b) f(x) = \cos(5x)\ at\ a = 2\pi \qquad (c) f(x) = \ln(x)\ at\ a = 1$$

6. For each of the following series, use the sequence of partial sums to determine whether the series converges or diverges

$$(a)\sum_{n=1}^{\infty} \frac{n}{n+2} \qquad (b)\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}$$